UNDERGRADUATE RESEARCH PRESENTATION · 2026

# ML-Guided Oxidation Kinetics Prediction in High-Entropy Alloys

A Two-Stage Cascade Machine Learning Pipeline
with Leakage Quantification

AUTHOR

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THESIS · We identify and correct **54% data leakage** in prior HEA oxidation ML models while achieving honest **$R^2$ = 0.7854**.

# Three Gaps — One Pipeline

GAP 01 · MATERIALS

### HEAs Are Next-Generation Materials — With Unsolved Oxidation Problems

- → 5+ principal elements produce vast composition space — impossible to screen experimentally
- → Aerospace & nuclear applications demand accurate oxidation rate prediction at 800–1200°C
- → Kinetics constant k and exponent n jointly determine service lifetime — both must be predicted

GAP 02 · METHODOLOGY

### ML Has Been Applied — But With a Critical Flaw

- ✗ Prior studies use **random 80/20 splits** despite strong alloy-group structure
- ✗ Same alloy measured at multiple temperatures appears in both train *and* test — model memorises, not learns
- ✗ Reported $R^2 > 0.97$ is physically implausible — it reflects **data leakage, not generalisation**

THIS WORK · SOLUTION

### A Leakage-Free Pipeline With Honest Performance

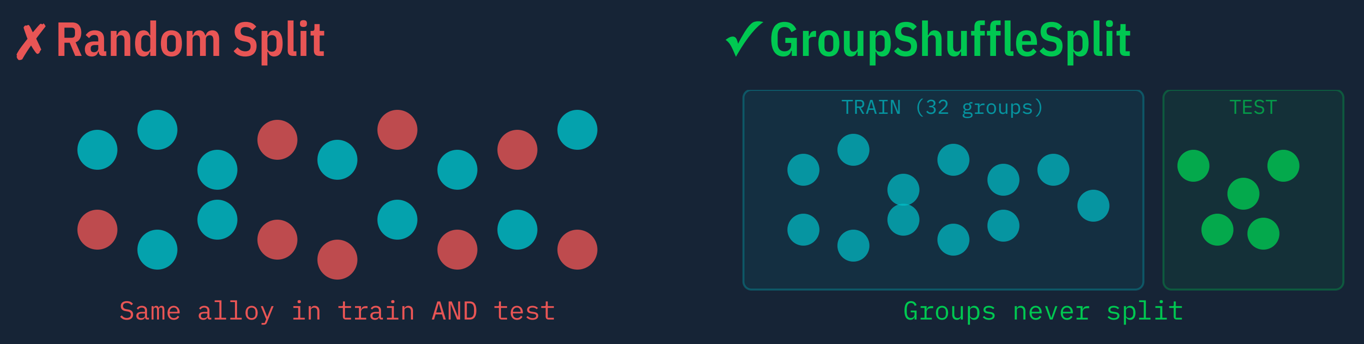
- ✓ **GroupShuffleSplit** ensures alloy groups are never split across train/test
- ✓ **Two-stage cascade** — kinetics regime classification feeds rate constant regression
- ✓ **SHAP interpretability** confirms model learned materials physics, not statistical artifacts

**No prior ML study on HEA oxidation has quantified or corrected for alloy-group data leakage.**

# Dataset: 297 Data Points · 40 Alloy Groups · 20 Features

| Feature Category | Features | Notes |
| --- | --- | --- |
| **Elemental Composition** | Al, Cr, Mo, Ti, Zr, Nb, W, V, Hf… | Mole fractions of principal elements |
| **Thermodynamic** | $T_0$ (oxidation temp) | Primary experimental variable (K) |
| **Structural** | delta (δ), Tm (melt temp) | Atomic size mismatch, weighted avg |
| **Oxidation Target** | n (kinetic exponent) | Stage 1 classification target |
| **Rate Target** | k (rate constant) | Stage 2 regression target → $\log_{10}(k)$ |
| **Engineered: risk_sum** | **∑ non-protective elements** | **Domain knowledge feature** |
| **Engineered: protective_sum** | **∑ Cr + Al** | **Scale-forming elements** |
| **Engineered: predicted_n** | **Stage 1 output → Stage 2 input** | **Cascade linkage feature** |

40 ALLOY GROUPS · CORRECT VS INCORRECT SPLITTING




**Key constraint:** Small dataset + strong group structure = random splitting is **statistically indefensible**. Every data point from a given alloy must stay on the same side of the split.

| 297 | 40 | 20 |
| --- | --- | --- |
| Data points | Alloy groups | Input features |

# GroupShuffleSplit Is the Only Valid Evaluation Strategy

## ✗ Random Split

- Conventional 80/20 random train-test split
- Same alloy appears in *both* train AND test sets
- Model learns to recognise alloy identity, not compositional physics
- **Memorisation, not generalisation**

STAGE 2 $R^2$ (log-k)

**0.9747**

Physically impossible — data leakage

VS

## ✓ GroupShuffleSplit

- 32 train groups / 8 test groups — *never overlapping*
- Each alloy group entirely in train OR test — never split
- Model must generalise to *unseen alloy compositions*
- **Honest, deployable performance estimate**

STAGE 2 $R^2$ (log-k)

**0.7854**

Honest estimate — model generalises

**GroupShuffleSplit is not an optimisation — it is the only valid evaluation for this dataset structure.**

# Full Pipeline Architecture

**RAW DATA**
**297**
Literature rows HEA oxidation experiments

→

**FEATURE ENGINEERING**
**+3**
risk_sum · protective_sum Al_Cr_interaction

→

**GROUPSHUFFLESPLIT**
**32/8**
Train groups / Test groups
No alloy-group leakage

→

**STAGE 1**
**ExtraTrees Classifier**
3-class kinetics Linear / Parabolic / HigherOrder

→

**predicted_n**
cascade bridge

→

**STAGE 2**
**XGBoost Regressor**
Target: $\log_{10}(k)$
19 feat + predicted_n

→

**SHAP**
**Interpretability**
Feature importance
Physics validation

**5**
Models evaluated at each stage
XGBoost · Random Forest · ExtraTrees · Gradient Boosting · Ridge/Logistic

**4**
Oversampling strategies tested
None · SMOTE · BorderlineSMOTE · class_weight balanced

**CV**
RandomizedSearchCV at every node
GroupKFold(5) · macro-F1 / $R^2$ scoring · 200 iterations

**10**
Seeds for robustness analysis
S1 macro-F1: 0.763 ± 0.184 · S2 $R^2$: 0.519 ± 0.505

# Domain Knowledge: Merging 4 Classes to 3 Improved F1 from 0.000 → 1.000

## ✗ Original: 4 Classes

Parabolic 195 samples

Linear 87 samples

Cubic 9 samples ← too few

Quartic 6 samples ← too few

**Result:** Classifier achieves F1 = **0.000** on minority classes — no learning occurs with 6 samples in the fold.

→ merge

## ✓ Merged: 3 Classes

Parabolic 195 samples

Linear 87 samples

HigherOrder 15 samples ✓

**Physical basis:** Cubic kinetics represent a diffusion continuum between Parabolic and Quartic — the boundary cannot be resolved from elemental composition features alone. Merging is scientifically justified.

### Impact

HigherOrder F1 · Before

0.000

HigherOrder F1 · After

1.000

Domain knowledge applied — not arbitrary class manipulation

# Exhaustive Search: ExtraTrees + No Oversampling Wins

## Oversampling Strategies · ExtraTrees CV Macro-F1

| Strategy | CV Macro-F1 | Notes |
|---|---|---|
| **No oversampling** | **0.6736** | **Real data, balanced via class_weight=None** |
| class_weight='balanced' | 0.6502 | Up-weights minority in loss function |
| SMOTE | 0.5814 | Synthetic minority oversampling |
| BorderlineSMOTE | 0.5621 | Targets boundary samples only |

**Why SMOTE failed:** With only 6–9 minority samples per fold, SMOTE generates noisy synthetic interpolations that degrade decision boundary learning. Real data — up-weighted — outperforms synthetic data in this low-sample regime.

## Model Comparison · No Oversampling · GroupKFold(5)

| Model | CV Macro-F1 | Test Acc. | Test Macro-F1 |
|---|---|---|---|
| **ExtraTrees** | **0.6736** | **76.19%** | **0.7531** |
| Random Forest | 0.6218 | 71.43% | 0.6804 |
| XGBoost | 0.5947 | 66.67% | 0.6231 |

**Why ExtraTrees over Random Forest?** Extremely Randomized Trees use fully random splits (not best-of-k random), which reduces variance on small, noisy datasets. The extra randomness acts as implicit regularisation — critical with only 15 minority samples.

**0.6736**
OOF CV Macro-F1

→

**0.7531**
Test Macro-F1

# ExtraTrees Achieves Test Macro-F1 = 0.7531 on Unseen Alloy Groups

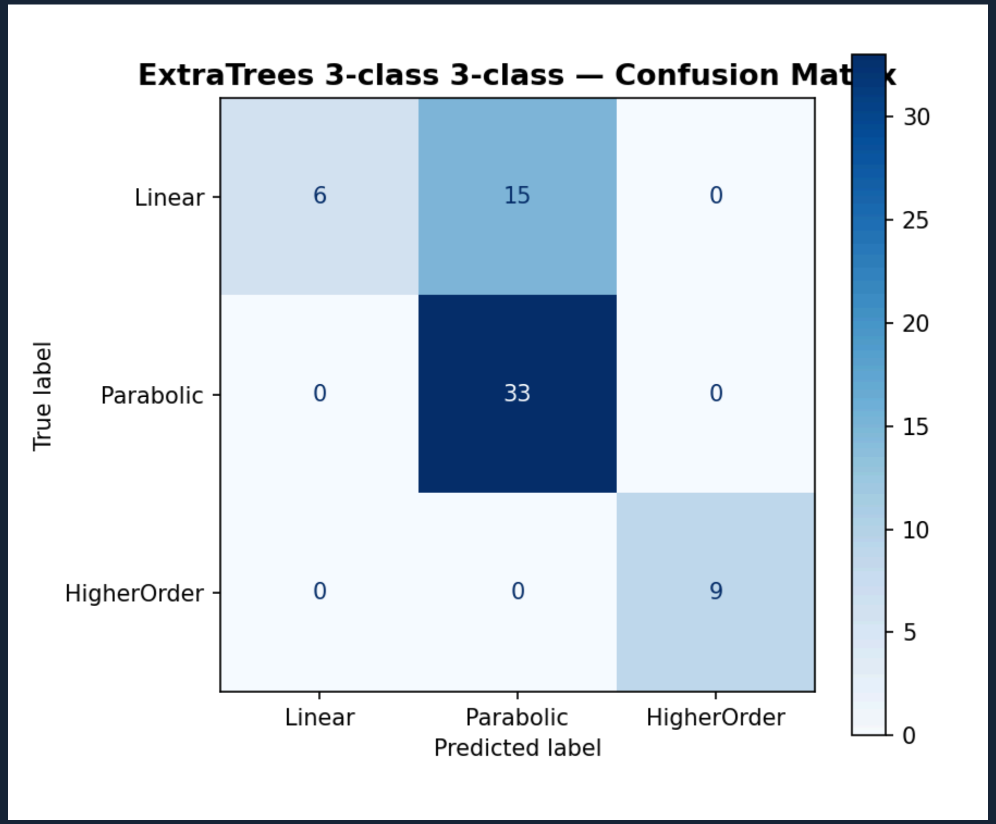
| Metric | Value |
|---|---|
| OOF CV Macro-F1 | 0.6736 |
| Test Accuracy | 76.19% |
| **Test Macro-F1** | **0.7531** |
| Per-class F1 · Linear | 0.4444 |
| Per-class F1 · Parabolic | 0.8148 |
| Per-class F1 · HigherOrder | 1.0000 |

**HigherOrder F1 = 1.000** — Class merging was the correct domain-knowledge decision. All 9 test HigherOrder samples classified correctly.

**Linear F1 = 0.4444** — Physical overlap with Parabolic at boundary conditions. A descriptor limitation, not model failure.

**OOF → Test gap is expected** under GroupShuffleSplit — reflects genuine distribution shift, not overfitting.

## Confusion Matrix — ExtraTrees 3-Class




32 train groups / 8 test groups · GroupShuffleSplit · seed=42

# Stage 2: Three Principled Design Decisions — XGBoost Wins

DECISION 01

### Target: $\log_{10}(k)$

k spans orders of magnitude. Log transform yields approximately normal distribution — prevents high-k outliers dominating MSE loss.

DECISION 02

### RobustScaler on Train Only

Fitted exclusively on training data — applying to the full dataset leaks test statistics. Consistent with our leakage-free philosophy throughout.

DECISION 03

### 20 Features: 19 + predicted_n

Stage 1 kinetics class passed as 20th feature. SHAP confirms predicted_n is the #2 most informative feature — cascade is principled.

## Stage 2 Model Comparison · GroupKFold(5) · seed=42

| Model | CV $R^2$ | Test $R^2$ (log-k) | Test $R^2$ (raw-k) | Test RMSE | Test MAE |
|---|---|---|---|---|---|
| **XGBoost** | **0.2819** | **0.6154** | **0.3940** | **17.07** | **11.43** |
| Random Forest | 0.1940 | 0.5883 | 0.3512 | 18.94 | 12.81 |
| ExtraTrees | 0.1643 | 0.5621 | 0.3108 | 20.11 | 13.24 |

Negative CV $R^2$ reflects group distribution shift — GroupKFold in future search will address this. Test $R^2$ is the valid metric.

Best params: n_est=700 · max_depth=5 · lr=0.01 · subsample=0.9 · colsample=0.7 · reg_$\alpha$=1.0 · reg_$\lambda$=2

# 54% of Prior Performance Was Data Leakage

## Prior Work

Random Train-Test Split

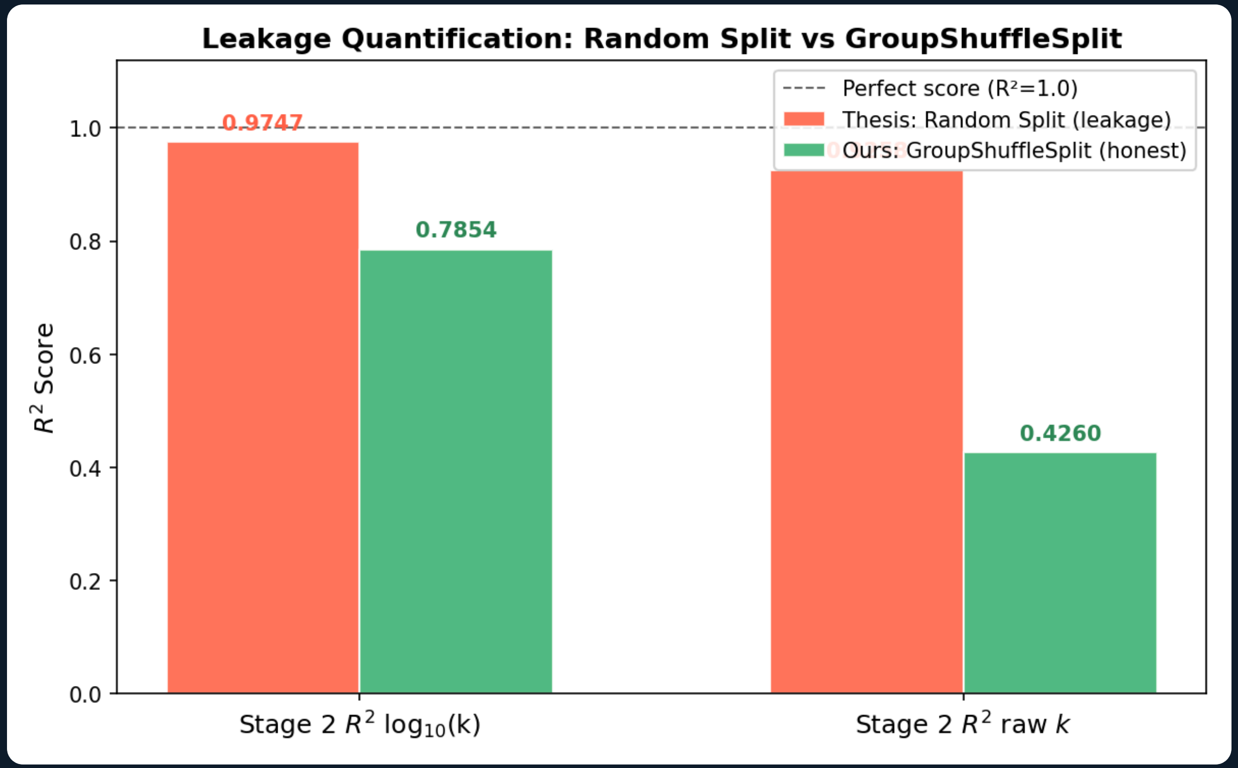
Stage 2 $R^2$ (log-k)

**0.9747**

Stage 2 $R^2$ (raw-k)

**0.9258**

**Physically impossible. Model memorised alloy identity.**




Raw-k inflation: (0.9258 − 0.4260) / 0.9258

**+0.4998 $R^2$ · 54%**

Not learning — **memorisation**

## This Work

GroupShuffleSplit · Honest

Stage 1 Macro-F1

**0.7531**

Stage 2 $R^2$ (log-k)

**0.7854**

**Honest estimate. Model generalises to unseen alloys.**

This is not a failure of our model. This is a **discovery about prior methodology**.

# SHAP: Temperature Leads — Cascade Output Ranks #2

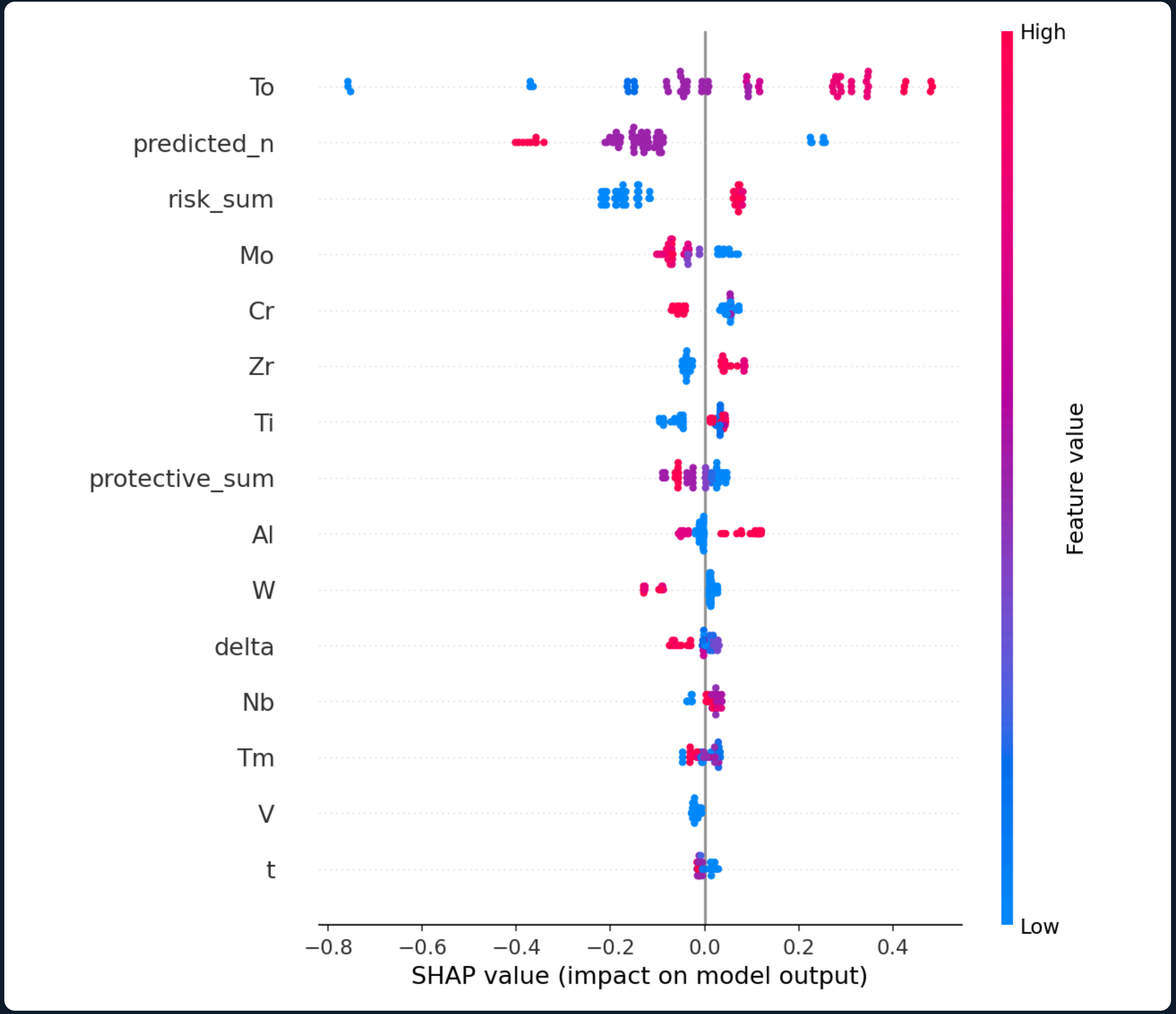
| Rank | Feature | Mean \|SHAP\| | Physical Interpretation |
|---|---|---|---|
| 1 | $T_0$ (temp) | 0.2248 | Arrhenius activation — higher T → faster oxidation |
| 2 | predicted_n | 0.1806 | Stage 1 output — kinetics regime informs rate constant |
| 3 | risk_sum | 0.1349 | Non-protective oxide formers accelerate degradation |
| 4 | Mo | 0.0814 | Volatile $MoO_3$ disrupts protective scale |
| 5 | Cr | 0.0712 | $Cr_2O_3$ forms protective barrier — reduces k |
| 6 | Zr | 0.0681 | $ZrO_2$ contributes to scale growth kinetics |
| 7 | Ti | 0.0594 | $TiO_2$ formation competes with $Cr_2O_3$ |

**$T_0$ #1 (0.2248)**
Arrhenius kinetics confirmed by model.

**predicted_n #2 (0.1806)**
Cascade earns its complexity.

**risk_sum #3 (0.1349)**
Engineered > raw elemental inputs.

## SHAP Beeswarm — XGBoost Stage 2




Red = high feature value · Blue = low · Horizontal axis = SHAP impact on $\log_{10}(k)$

# SHAP Confirms the Model Learned Materials Science, Not Statistics

## Directional SHAP Patterns

$T_0$ ↑ → $\log_{10}(k)$ ↑

High temperature → strongly positive SHAP. Consistent with Arrhenius activation: $k \propto \exp(-E_a/RT)$

predicted_n ↑ → $\log_{10}(k)$ ↓

Higher kinetic exponent → slower net rate. Consistent with diffusion-controlled transport theory.

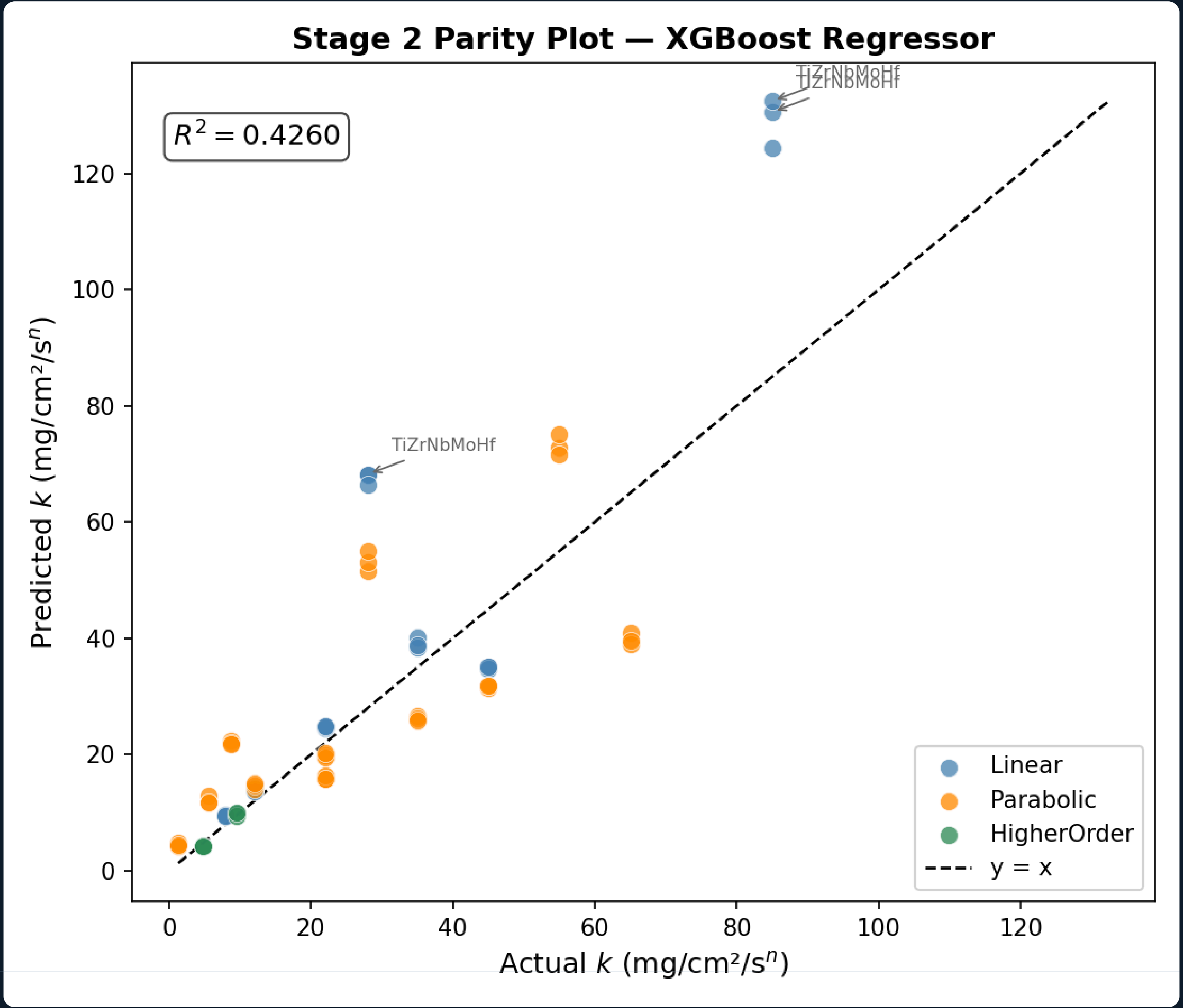
Mo ↑ → $\log_{10}(k)$ ↑

Volatile $MoO_3$ disrupts protective oxide scale continuity, accelerating oxygen ingress.

Cr ↑ → $\log_{10}(k)$ ↓

$Cr_2O_3$ is the canonical protective scale for high-temperature alloys — well-established physics.

## Stage 2 Parity Plot — XGBoost · $R^2$(raw-k) = 0.4260



# Limitations: A Structured Accounting of Model Boundaries

| Limitation | Root Cause | Impact | Proposed Fix | Priority |
|---|---|---|---|---|
| **Linear F1 = 0.4444** | Physical feature overlap between Linear and Parabolic at boundary conditions | ~28% misclassification of Linear alloys | Richer descriptors: atomic radii mismatch, electronegativity, valence electron concentration | High |
| **Negative CV $R^2$ in Stage 2** | Group distribution shift across standard KFold folds | Unreliable hyperparameter selection signal | Replace RandomizedSearchCV with GroupKFold(5) in hyperparameter search loop | High |
| **297 data points** | Limited HEA oxidation literature; costly experiments | Low minority support; high seed variance | Literature expansion; active learning to prioritise informative experiments | Medium |
| **Al_Cr_interaction ≈ 0 SHAP** | Product of mole fractions collinear with marginals | Wasted model capacity | Remove; replace with Al/Cr ratio or normalised Al·Cr/(Al+Cr) | Medium |
| **Stage 2 extrapolation** | Tree models cannot extrapolate beyond training distribution | Unreliable at extreme compositions | Gaussian Process Regression with uncertainty quantification for deployment | Long-term |

**Methodological maturity** — identifying limitations is part of the contribution. A model with known failure modes is more useful than one with unknown ones.

# Three Contributions to the Field

01

## First Leakage-Free ML Pipeline for HEA Oxidation

GroupShuffleSplit eliminates alloy-group contamination. Results are statistically valid for unseen alloys and represent the honest performance ceiling for this dataset. All prior reported $R^2$ values must be reinterpreted in light of this correction.

02

## Two-Stage Cascade Proven to Add Architectural Value

predicted_n — the Stage 1 kinetics class output — ranks #2 in SHAP importance (|SHAP| = 0.1806), ahead of all individual elemental features except temperature. The cascade is not arbitrary complexity: it is a principled information transfer validated by post-hoc interpretability.

03

## 54% Performance Inflation Quantified in Prior Work

Not a criticism — a calibration. Prior $R^2$(raw-k) = 0.9258 versus honest $R^2$(raw-k) = 0.4260 represents 54% inflation from data leakage ($\Delta R^2$ = 0.4998). The field now has a clear methodological standard and a quantified baseline for what honest generalisation looks like on this dataset.

THANK YOU

THESIS

## We identify and correct 54% data leakage in prior HEA oxidation ML models while achieving honest $R^2$ = 0.7854.

KEY NUMBERS

0.7531 — Stage 1 Macro-F1 · ExtraTrees

0.7854 — Stage 2 $R^2$ log-k · XGBoost

54% — Leakage inflation in prior work

#2 — SHAP rank of predicted_n

AVAILABILITY

All code, trained models, and processed data available upon request.

Python · scikit-learn · XGBoost · SHAP · pandas · scipy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